清华大学考试试题专用纸

姓名: _____ 学号: _____

Everyone needs to solve Problem 1, 2, 3, 4 (10 points each). Please select four problems in the following table and solve them (15 points each).

Problem 5	Problem 6	Problem 7	Problem 8	Problem 9	Problem 10

1. Construct two covering maps, $p : X \to Y$ and $q : Y \to Z$, among CW complexes, such that $q \circ p : X \to Z$ is not a covering map. Justify your answer.

2. Consider $\mathbb{RP}^n$, $n \geq 3$, with the standard metric induced from the standard sphere S^n of sectional curvature 1 via the antipodal quotient. Fix a point $[p] \in \mathbb{RP}^n$, where $p \in S^n$. Describe the cut locus $\mathrm{Cut}([p]) \subset \mathbb{RP}^n$ explicitly.

3. For each natural number $k \geq 0$, construct a closed connected 3-manifold such that

$$H_2(M, \mathbb{Z}) = \mathbb{Z}^k \oplus \mathbb{Z}/2.$$

4. Let (M^3, g) be a Riemannian 3-manifold and fix $p \in M$. Choose an orthonormal basis $\{e_1, e_2, e_3\}$ of T_pM that diagonalizes $\mathrm{Ric}_p : T_pM \to T_pM$ with eigenvalues $\lambda_1, \lambda_2, \lambda_3$.

   (a) Show that $\mathrm{sec}_p > 0$ if and only if $\lambda_i + \lambda_j > \lambda_k$ for all distinct i, j, k. In particular, $\mathrm{Ric}_p > 0$ (positive definite) does not imply $\mathrm{sec}_p > 0$.

   (b) Does $\mathrm{Ric}_p = 0$ imply $\mathrm{sec}_p = 0$ in dimension 3? Justify your answer.

5. Let $S \subset \mathbb{R}^3$ be a smoothly embedded closed surface. Prove that S is orientable.

6. Let (M, g) be a closed (compact, no boundary) Riemannian manifold with $\mathrm{Ric} \leq 0$ and let X be a Killing vector field. Show that if $\mathrm{Ric} < 0$ at some point of M, then $X \equiv 0$.

7. Suppose $f : \mathbb{C} \to \mathbb{C}$ is a complex polynomial in one variable. Let $F : \mathbb{CP}^1 \to \mathbb{CP}^1$ be the unique smooth map that is compatible with f. (Namely, let $i : \mathbb{C} \to \mathbb{CP}^1$ be the natural inclusion, then $F \circ i = i \circ f$.)

   Prove that the mapping degree of F equals the degree of f as a polynomial.

8. Let $\Omega = \mathbb{R}^n \setminus \{0\}$ and $u(r) = 1 + r^{2-n}$ (with $r = |x|$), $n \geq 3$. Consider the metric $g_u = u^{\frac{4}{n-2}}\,\delta$, where δ is the Euclidean metric. Decide whether (Ω, g_u) is complete, and justify your answer.

9. Let M be a closed oriented manifold of dimension n. Let $S \subset \mathrm{int}(M)$ be a compact embedded submanifold of dimension k. Prove that

$$H^{n-i}(S, \mathbb{Z}) \cong H_i(M, M \setminus S, \mathbb{Z}) \quad \text{for all } i.$$

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10. Let (Σ, g) be a Riemannian 2-dimensional surface and fix $p \in \Sigma$. For small $r > 0$, let $S_r(p) = \{q \in \Sigma \mid d(p,q) = r\}$ be the circle of radius r around p. Assume that there exists $r_0 > 0$ such that $\text{Length}(S_r(p)) \leq 2\pi r$ for all $r \in (0, r_0)$. Prove that the Gauss curvature at p satisfies $K_p \geq 0$.

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清华大学考试试题专用纸

考试课程 Subject: 求真书院代数博士生资格考 Algebra Qualifying Exam for Qiuzhen College

姓名 Name: _____

学号 Student ID: _____

- Exam Time: February 2026 (3 hours)
- This exam has 4 pages of 7 basic problems and 2 out of 6 advanced problems (100 points).
- Candidates are expected to adhere to the examination discipline by default, and those who do not comply will bear the consequences themselves.
- To receive full credit, all answers must provide necessary **details**, **proofs**, and **justifications**.
- You are strongly encouraged to write your answers **in English**.
- **第 8-13 题为提高题，你需要选择其中任意两题作答。请在答题纸首页清晰画出并填写下列表格，以注明需评分的两道题目。未在首页列出的题目将不予评分。**
  Q.8-13 are advanced problems and **selective**. Choose and answer **two** of them. No other answers will be graded. **Draw and fill the following table on the front page of the answer sheet to declare which two questions need to be graded.**

Advanced Problems		
Q.		
Score		

In the following problems, $\mathbf{Q}$ is the field of rational numbers, $\mathbf{R}$ is the field of real numbers, $\mathbf{C}$ is the field of complex numbers, $\mathbf{Z}$ is the ring of integers, $\mathbf{F}_q$ is the finite field consisting of q elements. **In all the problems, a ring means a commutative ring with multiplicative unit**.

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1 基础题 Basic Questions

Q. 1 (10 points). *Let $n \geq k \geq 1$ be positive integers. For a $\mathbf{C}$-linear endomorphism $f\colon V \to V$ of an n-dimensional $\mathbf{C}$-vector space, prove that the following are equivalent.*

(a) f is an isomorphism;

(b) the induced endomorphism $\otimes^k f\colon \bigotimes^k V \to \bigotimes^k V$ on the k-th tensor power is an isomorphism;

(c) the induced endomorphism $\wedge^k f\colon \bigwedge^k V \to \bigwedge^k V$ on the k-th exterior power is an isomorphism.

Q. 2 (10 points). *Let* $A = \begin{bmatrix} 3 & -2 & 1 \\ 1 & 0 & 1 \\ -1 & 2 & 0 \end{bmatrix}$.

1. *Find the eigenvalues of A.*
   Hint: one of the eigenvalues is an integer.

2. *Consider the set*

$$D = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbf{Q}^3 \;\middle|\; A^{100} \begin{bmatrix} x \\ y \\ z \end{bmatrix} - A \begin{bmatrix} x \\ y \\ z \end{bmatrix} - \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbf{Z}^3, \quad 0 \leq x, y, z < 1 \right\}.$$

   Find the cardinality of D.

Q. 3 (10 points).

1. *Show that the symmetric group S_4 has no normal Sylow subgroups.*

2. *Show that every group of order 24 with no normal Sylow subgroups must be isomorphic to S_4.*

Q. 4 (10 points). *Let $f(x) = x^5 - 11x - 1 \in \mathbf{Q}[x]$.*

1. *Show that $f(x)$ is irreducible over $\mathbf{Q}$.*

2. *Let G denote the Galois group of f. Show that G is isomorphic to the symmetric group S_5.*

Q. 5 (10 points). *Let R be a Noetherian local ring with maximal ideal $\mathfrak{m}$. Assume that $\mathfrak{m}^n = \mathfrak{m}^{n+1}$ for some $n > 0$. Show that R is Artinian.*

Q. 6 (10 points). *Consider the $\mathbf{Z}[x, y]$-algebras*

$$A = \mathbf{Z}[x, y]/(x^3 + xy + y^2 + 7x - 4, 10x - 5y + 1) \quad \textit{and} \quad B = \mathbf{Z}[x, y]/(2x - y - 1).$$

Set $C = A \otimes_{\mathbf{Z}[x,y]} B$. Let $f \in C$ denote the image of x and write C_f for the localization $S^{-1}C$ of C with respect to $S = \{f^n \mid n \geq 0\}$. Find all the elements $e \in C_f$ with $e^2 = e$.

Q. 7 (10 points). *Let G be a finite group. Let $\{x_i\}_{i \in I}$ be a set of representatives for the conjugacy classes of G, i.e., it contains exactly one element in each conjugacy class. Let $\chi : G \to \mathbf{C}^\times$ be an irreducible character of G. Write $n = \sum_{i \in I} \chi(x_i)$. Show that n is an integer greater or equal to zero.*

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2 高等题 Advanced Questions

Q. 8 (15 points). *Let t be a variable. Assume that there exist non-constant elements x, y in the field $\mathbf{C}(t)$ of rational functions such that*

$$y^n = \prod_{j=1}^{2026}(x-j)$$

for some $n \in \mathbf{Z}_{\geq 1}$. Show that $n = 1$.

Q. 9 (15 points). *Let K be an algebraically closed field and X the hypersurface in $\mathbf{P}^3_K$ defined by the homogeneous equation*

$$x^2w - y^2z = 0.$$

where (x, y, z, w) are homogeneous coordinates on $\mathbf{P}^3_K$. Prove that X is a rational variety, i.e., its function field is $K(t_1, t_2)$, where t_1, t_2 are algebraically independent over K.

Q. 10 (15 points). *Let $K = \mathbf{Q}(\sqrt{2}, \sqrt{3})$.*

1. *Find the ring $\mathcal{O}_K$ of integers of K.*
2. *Regard the unit group $\mathcal{O}_K^\times$ as a $\mathbf{Z}$-module. Find a $\mathbf{Q}$-basis of $\mathcal{O}_K^\times \otimes_{\mathbf{Z}} \mathbf{Q}$.*

Q. 11 (15 points).

1. *Let K be a number field and let $\mathcal{O}_K$ denote its ring of integers. Show that there exists a nonzero $f \in \mathcal{O}_K$ such that $\mathcal{O}_K[1/f]$ is a PID.*
2. *Let $K = \mathbf{Q}(\sqrt{-83})$. Prove or disprove the following statement.*

   There exists a prime $p \in \mathbf{Z}$ such that $\mathcal{O}_K[1/p]$ is a PID.

Q. 12 (15 points). *Consider the Lie algebra $\mathfrak{g} = \mathfrak{gl}_n(\mathbf{C})$. Let e_{ij} denote the elementary marix whose (i, j) entry is 1, and other entries are zero. Let $\mathfrak{b}$ be the Borel subalgebra consisting of upper-triangular matrices. Let $\mathfrak{t}$ be the Cartan subalgebra consisting of diagonal matrices. Let $V = \mathbf{C}^n$ be the standard representation of $\mathfrak{g}$. Write $\epsilon_1, ..., \epsilon_n$ for the weights of this representation. Recall that for each $\lambda \in \mathfrak{t}^*$, we have a Verma module $M(\lambda) = U(\mathfrak{g}) \otimes_{U(\mathfrak{b})} \mathbf{C}_\lambda$, where $\mathfrak{b}$ acts on $\mathbf{C}_\lambda$ through the quotient to $\mathfrak{t}$ and the latter acts by the weight λ.*

1. *Show that the module $M(\lambda) \otimes V$ admits a filtration whose successive quotients are Verma modules $M(\lambda + \epsilon_i)$ with i running over from 1 to n, each appearing exactly once.*
2. *Show that*

$$X : M(\lambda) \otimes V \to M(\lambda) \otimes V, \quad m \otimes v \mapsto \sum_{1 \leqslant i,j \leqslant n} e_{ij}(m) \otimes e_{ji}(v)$$

   is a $\mathfrak{g}$-module homomorphism.

   Hint: you might want to connect them to Casimir elements.)

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3. *Compute the generalised eigenvalues of X acting on $M(\lambda) \otimes V$. Show that each generalised eigenspace is again a $\mathfrak{g}$-representation filtered by Verma modules. Describe the successive quotients in this filtration for each possible eigenvalue.*

Q. 13 (15 points)**.** *Let (W, S) be a dihedral Coxeter group. That is*

$$W = \langle s, t \mid s^2 = t^2 = (st)^{m_{st}} = e \rangle$$

where $e \in W$ is the identity, and $m_{st} \in \{2, 3, 4, ..., \infty\}$.

1. *Let $\mathcal{H}$ be the Hecke algebra associated with the group (W, S). Write down the definition of $\mathcal{H}$ as a $\mathbf{Z}[v, v^{-1}]$-algebra in terms of generators and relations.*

2. *Write down the definition of the Kazhdan-Lusztig basis of $\mathcal{H}$. Denote elements in this basis by b_x for $x \in W$.*

3. *Given $0 \leqslant m \leqslant m_{st}$ write $st(m)$ for the element $stst...$ where m terms appear, and similar for $ts(m)$. For example $st(0) = e$, $ts(1) = t$, $st(2) = st$, $ts(3) = tst$, etc. Give explicit description of all elements in W and describe the Bruhat order on W explicitly.*

4. *Compute the expansion of b_x in terms of standard basis in $\mathcal{H}$ for all $x \in W$ and justify your answer.*

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Qualifying Exam-Analysis and Differential Equations

Spring 2026

1. (10 points) Compute the Fourier transform of the function $u(x) = \dfrac{1}{1+x^4} \in L^1(\mathbb{R})$, that is, for any $\xi \in \mathbb{R}$, compute the following integral

$$\hat{u}(\xi) = \int_{\mathbb{R}} \frac{e^{-ix\xi}}{1+x^4} dx.$$

2. (10 points) Find the solution to the initial value problem for $u = u(x)$,

$$x\frac{d^2u}{dx^2} - 2(x+1)\frac{du}{dx}x + (x+2)u = 0, \qquad u(1) = e, \;\; u'(1) = 2e.$$

(Hint: First check that $u = e^x$ is a solution to the equation.)

3. (10 points) Prove that the following limit exists in $\mathscr{D}'(\mathbb{R})$ (in the sense of distribution on $\mathbb{R}$), and find the limit:

$$\lim_{n\to+\infty} n^2|x|\cos(nx).$$

4. (10 points) Let X, Y be Banach spaces. Let $T : X \to Y$ be a bounded linear map. Suppose that T maps every bounded closed set to a closed set. Prove that $T(X)$ is closed in Y.

5. (15 points) Let $\Omega \subset \mathbb{C}$ be open and connected, and $f_n : \Omega \to \mathbb{C}$ be holomorphic (i.e., analytic) for each $n \in \mathbb{N}$. Suppose f_n converges to f uniformly on each compact subset of Ω, and f is non-constant. Let $U \subset \Omega$ be open. For each compact subset $K \subset f(U)$, prove that $K \subset f_n(U)$ for all sufficiently large n.

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6. (15 points) Let $f : \mathbb{R} \to \mathbb{R}$ be locally integrable (i.e. Lebesgue integrable on any compact interval). Assume that f is weakly differentiable, that is, its weak derivative f' (in the sense of distribution) is also a locally integrable function. Prove that, except for a zero measure set, f agrees with a function $g : \mathbb{R} \to \mathbb{R}$ that is absolutely continuous on any compact interval.

7. (15 points) Given $f \in C^1([0, \infty))$, consider the linear differential equation for $u = u(x)$ on $[0, \infty)$

$$\frac{d^2u}{dx^2} - (1 + f(x))u = 0.$$

   (1) (5 points) If $\lim_{x \to +\infty} f(x) = 0$, prove that any non-zero solution $u = u(x)$ has at most finite number of zeroes on $[0, \infty)$;

   (2) (10 points) If $\int_0^\infty |f(x)|dx < \infty$, prove that there is a unique solution $u = u(x)$ satisfying

$$\lim_{x \to +\infty} e^x u(x) = 1.$$

8. (15 points) Let $\mathcal{H}$ be an infinite-dimensional separable complex Hilbert space. Consider the following two questions:

   (1) (8 points) Let T be a bounded linear operator on $\mathcal{H}$. Suppose that for any orthonormal basis $\{f_n\}_{n=1}^\infty$ in $\mathcal{H}$ we have $\lim_{n \to \infty} Tf_n = 0$. Prove that T is a compact operator. (Hint: Approximate T by finite-rank operators defined using a fixed orthonormal basis.)

   (2) (7 points) Let $\mathcal{H} = l^2(\mathbb{Z}_+)$. Prove that there is a bounded linear operator T on $l^2(\mathbb{Z}_+)$ such that

$$\begin{aligned} &T(a_1, a_2, a_3, \dots) \\ &= \left(a_1, \frac{1}{\sqrt{2}}(a_2 + a_3), \frac{1}{\sqrt{3}}(a_4 + a_5 + a_6), \frac{1}{\sqrt{4}}(a_7 + a_8 + a_9 + a_{10}), \dots\right) \end{aligned}$$

   holds if $\{a_n\}$ contains only finitely many non-zero terms, and that T is not a compact operator. (Hint: Consider the adjoint of T.)

   Note: Part (2) shows that the condition "for every orthonormal basis" in (1) cannot be weakened to "for some orthonormal basis".

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